

MechWork.Rmd

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Introduction

This file investigates questions about world aggregate mechanical work. The questions were raised by Tiago Domingos regarding Figure 7 of the paper P. E. Brockway, M. K. Heun, Z. Marshall, E. Aramendia, P. Steenwyk, T. Relph, M. Widjanarko, J. J. Kim, A. Sainju, and J. Irtube. A country-level primary-final-useful (CL-PFU) energy and exergy database: Overview of its construction and 1971–2020 world-level efficiency results. Environmental Research: Energy, 1(025005):1–29, jun 2024.

Data

Gather and save data from the database.

```
conn <- PFUPipelineTools::get_mexerdb_conn(user = "dbcreator")
world_mats <- PFUPipelineTools::pl_filter_collect(
  db_table_name = "PSUTReAllChopAllDsAllGrAll",
  Country == "World",
  conn = conn,
  collect = TRUE)
world_mats |>
  saveRDS(file = "world_mats.rds")
```

Analysis

Load data.

```
world_mats <- readRDS(file = "world_mats.rds")
```

```
# world_mats_filtered <- world_mats |>
#   # Filter to the same parameters for Figure 7.
#   dplyr::filter(Dataset == "CL-PFU IEA+MW",
#                 LastStage == "Useful",
#                 IncludesNEU == FALSE) |>
#   dplyr::select(-R, -U, -U_EIOU, -U_feed, -r_EIOU, -V, -S_units) |>
#   dplyr::mutate(
#     ProductAggregation = factor(ProductAggregation, levels = c("Specified", "Despecified", "Grouped")),
#     IndustryAggregation = factor(IndustryAggregation, levels = c("Specified", "Despecified", "Grouped")),
#   ) |>
#   dplyr::arrange(Year, ProductAggregation, IndustryAggregation)
```

Make a graph similar to Figure 7.

```
world_mats |>
  # Filter to the same parameters for Figure 7.
  dplyr::filter(Dataset == "CL-PFU IEA+MW",
                Year >= 1971,
```

```

        LastStage == "Useful",
        IncludesNEU == FALSE,
        ProductAggregation == "Grouped",
        IndustryAggregation == "Grouped") |>
dplyr::arrange(Year, EnergyType) |>
dplyr::select(-R, -U, -U_EIOU, -U_feed, -r_EIOU, -V, -S_units) |>
dplyr::mutate(
  ProductAggregation = factor(ProductAggregation, levels = c("Specified", "Despecified", "Grouped")),
  IndustryAggregation = factor(IndustryAggregation, levels = c("Specified", "Despecified", "Grouped")),
  Y = Y |>
    matsbyname::select_cols_byname(
      remove_pattern = RCLabels::make_or_pattern(c("Exports",
        "Exports to World aviation bunkers",
        "Exports to World marine bunkers",
        "Losses",
        "Statistical differences",
        "Stock changes"))) |>
    matsbyname::select_rows_byname(remove_pattern = "NEU") |>
    matsbyname::rowsums_byname() |>
    matsbyname::hadamardproduct_byname(1e-6) # Convert from TJ to EJ
) |>
tidyr::pivot_longer(cols = "Y", names_to = "matnames", values_to = "matvals") |>
matsindf::expand_to_tidy(drop = 0) |>
dplyr::mutate(
  rownames = factor(rownames, levels = c("IP", "L", "HTH", "MTH", "LTH",
    "LTC", "HPF", "Prop", "MechW", "HuMech", "AnMech"))
) |>
ggplot2::ggplot(mapping = ggplot2::aes(x = Year,
  y = matvals,
  fill = rownames)) +
ggplot2::geom_area() +
ggplot2::facet_grid(cols = ggplot2::vars(EnergyType)) +
ggplot2::scale_x_continuous(breaks = c(1980, 2000, 2020)) +
ggplot2::scale_fill_manual(values = useful_carrier_colours) +
ggplot2::labs(x = NULL, fill = NULL, y = "Useful energy and exergy [EJ]") +
MKHthemes::xy_theme()

```

```
## Loading required namespace: Matrix
```

